Lexical_1

- Welcome & technical issues & questions
- How we will work together & grading

Regular expressions & automata

- Recap on regular expressions and automata
- Which regular expressions match which words?
  Regular expressions:

  ```
  "", a, ab, a|b, a*, a|b*, ab*, a*b, a*b*, a*|b*, (ab)*
  ```

  Words:

  ```
  ε, a, b, aa, ab, ba, aaa, aab, aba, abb, baa, bab, bba, bbb, aaaaaa, bbbbb, ababab, babab
  ```

- A tool that we will use today: **https://regex101.com/ (https://regex101.com/) (https://regex101.com/)**

Exercises

Create regular expressions for the following.

- Numbers:
  - integral (has at least one digit between 0 and 9)
  - such an integer that can not start with leading zeros (except the 0 itself)
  - signed integer (optionally having a + or a - sign as a first character)
  - fractional number with a decimal point (having at least one digit before the decimal point)
- Identifier: starts with a non-capital letter; after that, it can contain any number of letters (both non-capital and capital), digits, and an _ (underscore) sign
  - version2: the last character can not be an _ (underscore)
  - version3: two _ (underscore) can not be next to each other (a_a_a is valid while a__a is not)
- A more complex identifier:
  - the first part of the identifier can only contain letters and _ (underscore), while the second part can only contain digits.
  - version2: if the first part starts and also ends with _ (underscore), then it can only contain capital letters; otherwise, only non-capital letters
  - version3: if the second part - which can only contain digits - starts with an even digit, it is only accepted to have an even number of digits after it; if the first one is odd, then it is only accepted to have an odd number of digits (so the restriction is not on the digits themselves but on their total number)
- Recognize time in hh:mm format, where hh can be 00..23, while mm can be 00..59.
- Line comment, starting with // and ending at the end of the line

- Multiline comment, starting with /* and ending with the first */
  - version1: the inside of the comment can only contain letters
  - version2: the inside of the comment can contain anything except the * (star) sign
  - version3: even the * (star) is accepted inside the comment